The FDA has proposed the regulation of human cells, tissues, and cellular and tissue based products (HCT/Ps) to prevent the introduction, transmission, and spread of communicable diseases; however, these regulations infringe on human rights and stall innovation in the stem cell research industry. Additional time has been given from the original proposed date of the regulatory hearing because there is considerable interest in this topic and a need for stakeholder comments. Groups affected by this regulation include, but are not limited to: tissue establishments, biological and device manufacturers, health care professionals, clinicians, biomedical researchers, and the public. Comments requested from this group should be directed towards the scope, topics, questions, additional issues that need guidance, and the clarity of the regulations. The FDA has also scheduled a scientific workshop to identify and discuss the scientific considerations and the challenges of HCT/Ps in the market (regulations.gov).

HCT/Ps are articles consisting of human cells or tissues that are intended for implantation, transplantation, or transfer into a human. HCT/Ps are currently regulated under Section 361 of the Public Health Service Act, originally intended to prevent the spread of diseases from foreign countries. The Public Health Service Act also established the government's authority to quarantine individuals with harmful communicable diseases. HCT/Ps are regulated if they meet certain criteria. If they are: minimally manipulated, intended for [1]homologous use, there is no combination of cells or tissue with another article (except water, crystalloids, or a

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[1] For HCT/Ps to be intended for homologous use, they need to be repurposed for the same function that they originally existed.

sterilizing or preserving agent), they do not have a [2]systemic effect and is not dependent on the metabolic activity of living cells, they do have systemic effect and depend on metabolic activity of living cells and are used [3]autologous-ly, they are under regulation. If HCT/Ps don't follow under these criterion, they are regulated as a drug, device, or biological product under the Federal Food, Drug, and Cosmetic Act. The FDA has issued four draft guidances to further explain the criteria of the HCT/Ps to be regulated including the same-surgical-procedure exception, the minimal manipulation of HCT/PS, HCT/Ps from [4]Adipose tissue and the homologous use of HCT/Ps.

Based on the comments received on this proposed regulation, this regulation will hurt more people than it will help. One person commented about his stem cell adipose tissue replacement, and told how it "gave energy to move across the country six months later and helped with walking." With this regulation, this man wouldn't have the ability to have a stem cell adipose tissue replacement, and other various rights would be stripped. With this regulation, the use of one's own stem cells would be limited, and people would not have the ability to have their *own* stem cells frozen for continued use after a one year period, the stem cells would have to be thrown away within 24 hours. If someone has the ability and wants to repurpose their stem cells for future rehabilitation, why shouldn't they be allowed to?

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[2] For HCT/Ps to be systemic, they depend on the rest of the body's functions to survive and function. To be non-systemic, HCT/Ps do not depend on the function of anything else to survive and function.

[3] For HCT/Ps to be considered for autologous purposes, they can only be reused in the same body they were taken from.

[4] HCT/Ps from Adipose tissue are tissues used for storage or fat.

Stem cell research is still fairly new, and many people undergoing the surgeries are undertaking the risks and complications that are currently unknown. However, those that have undergone the surgeries have led better quality lives that they wouldn't be able to without the "experimental" surgeries. The more people research and attempt these surgeries, the more innovative and better the market will be for stem cell use. Unfortunately, the FDA has taken a "risk-based" approach on this issue, when there already isn't enough information to determine whether the benefits outweigh the risks, or if stem cell replacement use is truly dangerous. Their approach will limit the ability of people able to get these surgeries, even when they are paying out-of-pocket and have fully consented to the risks. Because the FDA has taken this approach and wants to create new guidances and regulations on stem cell treatment and study, it will drive people to look for these beneficial treatments elsewhere. Furthermore, it will make these potentially useful therapies into harmful practices underground or unregulated outside of the country—or under false cosmetic names. By regulating this potentially revolutionary and helpful industry, the FDA will most likely be making a Type II error, by regulating stem cell use and calling it unsafe, when in reality, it is safe.

A commenter with Parkinson's Disease revealed that he can get stem cell treatment outside of the US, and even right over the border, but decided to have his treatment in the US using the currently legal procedure with adipose tissue. The commenter stayed for treatment in the US when he knew that the out of country legal procedure using the tissue from a fetus or placenta would better cure him; however, that option is illegal in the US. Although this commenter stayed in the US, by regulating stem cell use in the US, people are more likely to go outside of the country to receive treatment—further investing money into other countries and allowing them to innovate while the US remains stagnant. With the FDA's proposed regulation, cutting edge treatment will be driven out of the country, further making it only available to the rich to receive safe treatment out of the US. For the rest that can't afford to travel and their insurance won't cover the cost of the regulated surgery will only have the option to turn to unregulated and potentially unsafe procedures.

The government's role in stem cell treatment should be to protect the consumer, not limit them. If the FDA took a different approach in order to protect the consumer by informing them, conducting tests, surveys, and scientific research, industry professionals would strive to use the best medical science. Instead, the opposite effect is happening because innovation will be stopped and current treatments won't be done in the best facilities or under the best care and attention. By regulating the procedures to be out of reach to the public, the adverse effects are far greater than allowing the industry to first innovate and then determine which regulations are appropriate. By stripping researchers the right to potentially help people and to improve stem cell use, public information about the use of stem cells will be limited. Patients that only have the option of using stem cell treatments will not be able to evaluate the benefits and risks of undergoing such treatments, and they will turn to unregulated, untested procedures. The FDA could play an important role in the improvement and innovation of the industry, that could potentially save many peoples lives that struggle from different conditions. Stem cell therapy can offer great benefits to patients whose options are limited or non-existent; the proposed regulations will take away these patients last resort options at a better quality of life.

People struggling with incurable problems are willing to undergo these new and experimental stem-cell procedures that have unproven therapies. With the current cutting edge research; however, if these regulations are imposed, the development of treatments will not be possible—which is a detriment to all. Currently, the possibilities of stem cell use are infinite. People are finding hundreds of ways to use their own stem cells, others', or freezing their's for the future to help cure, or alleviate, many diseases and ailments. To protect the interest of patients, and to support the innovation of this rapidly developing field, the FDA needs to facilitate the improvement of the information available and provide the people with the facts, so that they can make decisions about the uses of their *own* bodies. What justifications does the FDA have to prevent people from using their bodies as they wish, and for their own betterment, and with full knowledge of the risks they are taking? Adam Thierer in his book *Permissionless Innovation*, perfectly states the necessary role government should have with their regulations:

> "…problems will develop. But how these concerns are dealt with matters deeply. We should exhaust all other potential non regulatory remedies first before resorting to… controls on new forms of innovation…solutions should generally trump…controls… Companies, advocacy groups, and the government should all work together to educate consumers about proper use and corporate and personally responsibility to head off those problems to the maximum extent possible…" (103, Thierer).

It is known, across all industries and practices, that problems are bound to happen while innovation is present and happening. Before regulating based off of "what if" concerns and taking too many precautions, the FDA needs to allow the industry to develop and grow in order to make the correct decisions for everyone. If every industry was regulated based off of this approach, innovation would not move fast enough in the US and incentive to innovate would decrease. However, the FDA does have the option to regulate once they have conducted the necessary experiments and has assessed the situation to make it fair for all. And, if groups involved stem cell research and other industries all work together, they can truly better the options available to people while maximizing potential innovation in industries.

In George Stigler's *Theory of Economic Regulation,* he states that because the government is able to regulate the market, they have the power to help or hurt certain industries. In this case, it is possible that the FDA is helping the pharmaceutical industry and hurting the biomedical research industry, which affects the public as a whole. While the government can mandate regulations, sometimes regulations are sought out in certain industries (3, Stigler). The pharmaceutical industry could be seeking regulation to prevent the development of stem cell research. When an industry, like pharmaceuticals, receives power from a regulation of the of others, the costs of the community outweigh the benefits to the one industry (10, Stigler). The industry benefits more than the people the regulation affects because the political decision does not take in to account the costs that are distributed to the people. Stem cell science, which has the key to potential cures and a better quality of life, is prolonged by Big Pharma. The pharmaceutical industry would lose many customers and profits if stem cell research was allowed to continue to find cures for conditions and diseases that require billions of dollars for medications every year. As a governing agency, the FDA *should* have the best interest of the individual in mind, unfortunately, it is not the case when large amounts of money and pressure from the pharmaceutical industry are present. The FDA should be promoting the passionate and educated scientists that want to bring these treatments to people, but the persuasions of Big Pharma outweigh these decisions because of bribery. Big Pharma has the ability to persuade and guide these regulations through their trillion dollar industry and delay the discoveries of the possible uses of stem cell treatments. For those that have become successful in the free market, they hope to keep the market where it was when they were successful and not continue its ever-changing cycle. The pharmaceutical industry wants to maintain the time in the market where their profit margins were the greatest, which is a time when innovation is not happening. A commenter with Multiple Sclerosis commented on the FDA's docket and has a compelling argument about the regulations of the FDA, and how the pharmaceutical industry can benefit from regulations:

> "Dear FDA and all others who are in charge of my health, I…am on drugs that cost over 60,000 a year. They won't cure me, in fact I am told I will get worse. I would do stem cell in a minute…I would leave the country if I could afford it. I would do whatever it takes if I could just get my legs back where they were. To walk a mile with my daughter, get up with out spastic legs. To be able to focus on my work instead of fighting to have the right to do what I want to try and get healthy…If I want to to use my own cells or bone marrow or whatever to help myself why won't you let me? We just want to be healthy. Our bodies are our bodies if we can benefit and repair ourselves with ourselves why would you want to control that and stop it?" (regulations.gov)

In many instances, the only options people have to possibly help them is to buy into the pharmaceutical industry. Drugs are prescribed that only alleviate some pain but don't cure, or the prescriptions only prolong the illness. Even then, these semi-helpful options cost the patient thousands of dollars each year. The FDA doesn't seem to have much standing in this issue, or logical reasoning and intent to prohibit people from using their own stem cells to avoid the added costs and repercussions of prescriptions. If their intent is to prevent the spread of communicable diseases, they should become more informed first of the potential risks, rather than taking the risk-based approach that is costing many the ability to have a enjoyable life. If the FDA is being persuaded by the pharmaceutical industry to instill regulations on innovative stem cell research they are doing a dis-service as an agency that is supposed to protect people. In Lawrence Lessig's article, "Capitalism's Biggest Enemies: Elites Who Advocate Free-Market Competition," he quotes Bessen saying:

> "Government policies have…increasingly favored powerful interest groups at the expense of promising young start-ups stifling technological innovation."

Stem cell research has incredible potential to help so many people, and the tendency for the government to favor profit over the interests of the population is hurting the potential of the stem cell research industry.

While the FDA can be persuaded by powerful interests, their ability to enact appropriate regulation also suffers from time consuming government procedures and required processes. According to Milton and Rose Friedman in *Who Protects the Consumer?,* the FDA does have beneficial roles in ensuring products are safe to consume and use; however, it prolongs the process for drugs and services to be available on the market—perhaps for someone who needs it immediatley (204, Friedman). If cutting edge treatments are available, but have not been tested thoroughly yet, they might not become available until the FDA can ensure their success. For someone that needs treatment available, they might not be able to get it in the US because of the FDA's regulatory procedures. The FDA also errs on the side of caution because the government has taken on the risk of regulating this industry, and by doing so, they assume incredible liability for their decisions. If the FDA can take their time in reviewing a product or service, they will, because no one will know that this potentially beneficial service or product could be available. On the other hand, if the FDA rushes to approve an item and it is dangerous, they are branded with reckless decision making. It it therefore in their best interest—and not in the best interest of the population—to take their time in approving every drug or practice to avoid approving a dangerous drug or practice. However, the FDA approaches issues slowly because they are confronted with inefficient government practices and limitations. FDA regulations in this case, although meant to be good, suffer from being too cautious and having slow review processes.

The FDA needs to reconsider passing this regulation and allowing the stem cell research industry the ability to develop and assess the risks of its practices. The use of HCT/Ps has the ability to provide many temporary to permanent relief, and a return to a normal and functioning life. It was upsetting reading the comments on regulations.gov and seeing people pleading for this regulation not to pass. Stem cell use, for many, is a last resort for incurable diseases and conditions. And, as it seems, stem cell practices have so far been successful and have provided many people temporary to long term relief to their ailments. For people to not be allowed the use of their own body for betterment infringes on personal rights and freedoms. People that have undergone HCT/P related procedures have been fully consenting and aware that the potential risks are unknown; therefore, if they are consenting to "experimental" procedures, they should be allowed to.

It is unfair for the FDA to provide regulations before this industry has had the time to develop and research on the costs and benefits of stem cell use and practices. The FDA didn't seem to have a compelling case on exactly why stem cell use should be limited, and they did not have the facts to back up their claims on the hope to prevent the spread of communicable diseases by limiting the use of stem cell treatments. If people aren't allowed to have access to the treatments they need in the US they will find it elsewhere. By imposing this regulation it is having the opposite affect of keeping people safe, they will be forced to undergo unsafe and unregulated practices in the US or find it outside the borders. If people find their treatments elsewhere, it creates more danger for the spread of foreign and communicable diseases—which is what the FDA is trying to prevent in the first place.

If the FDA approached this situation as an opportunity to research, innovate, and inform the population, people would be better off. Because the patients seeking these treatments are consenting individuals, they would benefit from innovation in the industry and the availability of studies, research, and available information on their options. The ability to experiment in this field will incentive researchers and medical professionals to find the cures and options for those seeking them.

Sources Referenced

Friedman, Milton, Roy Hattersley, Peter Jay, Charles Medawar, and Saxon Tate. Who Protects the Consumer? N.p.: n.p., 1980. Print.

Stigler, George. The Theory of Economic Regulation. N.p.: n.p., n.d. Print.

Thierer, Adam. "Book - Permissionless Innovation." Permissionless Innovation. N.p., n.d. Web. 25 July 2016.

Comments on regulations.gov